

Mohammad Saad Nathani

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Applied ML/DL engineering student with hands-on experience across deep learning, RAG systems, and agentic AI tooling.
ACM Student Chapter Chair, NMIMS Shirpur.

Education

SVKM NMIMS Mukesh Patel School of Technology Management and Engineering	2023 – 2027
Bachelor of Technology in Artificial Intelligence and Machine Learning	CGPA: 3.62 /4.00 (6th Sem)
St. Xavier's High School, Vapi	10+2

Experience

CN Automation LLP [Link]	Jun 2026 – Jul 2026
<i>Engineering Intern (On-site)</i>	
- Designed and deployed a local RAG system for industrial manuals using Docling, Ollama (qwen2.5:7b, nomic-embed-text), and ChromaDB, containerized with Docker with a Streamlit front end for internal use. - Built a real-time Node-RED dashboard for OPC-UA plant KPIs and implemented PLC automation logic, including a TIA Portal Openness MCP server in Siemens TIA Portal.	

Projects

- ESG Survey Dashboard – Mirae Asset Foundation [Link]	Mar 2026
<i>Python, Power BI</i>	
Built the data pipeline and custom ESG scoring logic behind a survey analytics dashboard, processing 1,641 responses across 21 villages in Shirpur district; the scoring system classified the district in the "Safe" ESG category, indicating acceptable compliance.	
- ShopSense AI – AI-Powered Shopping Assistant	Jul 2025
<i>Python, OpenAI API, Llama 7B (local), Chrome Extension APIs</i>	
Built a browser extension that scrapes and structures product data from e-commerce pages, using an LLM to answer natural-language shopping questions compatibility, use-case fit, and recommendations across categories like tech and skincare. Supported both OpenAI's API and a locally-hosted Llama 7B model, giving users a free/offline option without requiring an API key.	
- FloatChat – Natural Language Climate Data Query System	Aug 2025
<i>Python, FAISS, PostgreSQL, NetCDF/Xarray, Claude, LangChain, MCP</i>	
Architected and led an end-to-end pipeline for ingesting daily Argo NetCDF ocean datasets into PostgreSQL; developed an MCP server exposing the data to an LLM agent for natural-language queries, rendering results as interactive maps and replacing manual report generation.	
- Holistically-Nested Edge Detection [Link]	Oct 2025
<i>TensorFlow, VGG16, OpenCV</i>	
Rebuilt the HED architecture from Caffe to TensorFlow, achieving an ODS F-score of 0.782 on BSD500 and 60% faster GPU inference compared to the original Caffe implementation. Packaged and published the model as a pip-installable Python library on PyPI for public use.	

Capstone Project

Surveillance Anomaly Detection with VL-JEPA	Ongoing
<i>PyTorch, Hugging Face</i>	
Developing a surveillance anomaly detection system using the VL-JEPA architecture, training on annotated UCF-Crime data to detect anomalies and generate natural-language alerts for authorities — designed with edge-device constraints in mind. (team capstone, 15-week timeline).	

Achievements

• 1st Place — AISEhack Theme 2: Pollution Prediction (ANRF, IBM, IITD, IIITH) [Link]	Apr 2026
• 2nd Place — Yugantar 2025 (Smart India Hackathon Internal Round) [Link]	Sept 2025
• Remote Finalist — HackRx 6.0 (Bajaj Finserv Hackathon 2025)	Aug 2025

Skills

Languages: Python, C++
AI/ML Core: PyTorch, LangChain, LangGraph RAG,MCP, Claude SDK, OpenAI
Libraries: NumPy, Pandas, Seaborn, Matplotlib, Plotly, OpenCV, NLTK, Scikit-learn, XGBoost
Frameworks/Tools: Streamlit, FastAPI
Data/BI: MySQL, Oracle SQL, Tableau, Power BI, SSIS, SAS

Certifications

Machine Learning - Banaras Hindu University (Score: 90%) [Link]	May 2025
Model Context Protocol: Advanced Topics - Anthropic [Link]	July 2026
Excel: Beginner to Advanced - Udemy [Link]	Dec 2024

Leadership & Extracurricular

Chair, ACM Student Chapter – NMIMS MPSTME Shirpur [Link]	Oct 2025
Led chapter operations and organized technical events, workshops, and student engagement initiatives.	